

Discovering the Metric Linkage with HSIC

a minimal prototype

1 Idea

Business metrics live at different levels (order, user, merchant) and have *no explicit formula* linking them. We only want to know *which metrics move together* — the association *linkage* — not any causal direction. The Hilbert–Schmidt Independence Criterion (HSIC) is a kernel measure of statistical dependence that is zero iff two variables are independent and, unlike correlation, detects *non-linear* association without assuming a functional form. Running HSIC on every pair of metrics reveals the linked chain and isolates the unrelated ones.

2 HSIC

For paired samples $\{(a_i, b_i)\}_{i=1}^n$, with RBF kernels $k(a, a') = \exp(-\|a - a'\|^2 / \sigma_a^2)$ and $l(b, b') = \exp(-\|b - b'\|^2 / \sigma_b^2)$ (bandwidths by the median heuristic), form Gram matrices $K, L \in \mathbb{R}^{n \times n}$ and the centering matrix $H = I_n - \frac{1}{n} \mathbf{1}\mathbf{1}^\top$. The empirical HSIC is

$$\widehat{\text{HSIC}}(a, b) = \frac{1}{(n-1)^2} \text{tr}(KHLH) \geq 0, \quad (1)$$

large when a, b are dependent and ≈ 0 when independent. It is symmetric and makes *no* directional / causal claim — it is purely an association.

3 Method

1. Compute each metric per entity (a common index), giving columns $m_1, \dots, m_p$.
2. For every pair compute $\widehat{\text{HSIC}}(m_i, m_j)$.
3. Keep pairs with $\widehat{\text{HSIC}}$ above a small threshold as *linkage edges*; the connected component is the metric *chain*.

4 Result

On a synthetic marketplace with a linked chain `delivery_mins` – `user_rating` – `n_orders` – `gmv` (metrics across the order, user, and merchant levels) plus two unrelated metrics (`quantity`, `discount`), the pairwise HSIC is:

Takeaway. HSIC alone — one symmetric, assumption-light, kernel dependence number per pair — is enough to *discover the metric linkage across levels*. Direction and root cause are deliberately *not* claimed here; this prototype only recovers which metrics are associated.

metric a	metric b	$\widehat{\text{HSIC}}$
delivery_mins	user_rating	0.055
delivery_mins	n_orders	0.056
delivery_mins	gmV	0.046
user_rating	n_orders	0.062
user_rating	gmV	0.054
n_orders	gmV	0.070
<i>any pair involving quantity or discount</i>		≤ 0.002

Table 1: Pairwise HSIC. The four chain metrics are mutually dependent ($\widehat{\text{HSIC}} \in [0.046, 0.070]$); every pair touching `quantity` or `discount` is ≤ 0.002 (independent). HSIC recovers the linkage chain `delivery_mins`–`user_rating`–`n_orders`–`gmV` and isolates the unrelated metrics — with no explicit relationship supplied and no causal direction claimed.